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State model…

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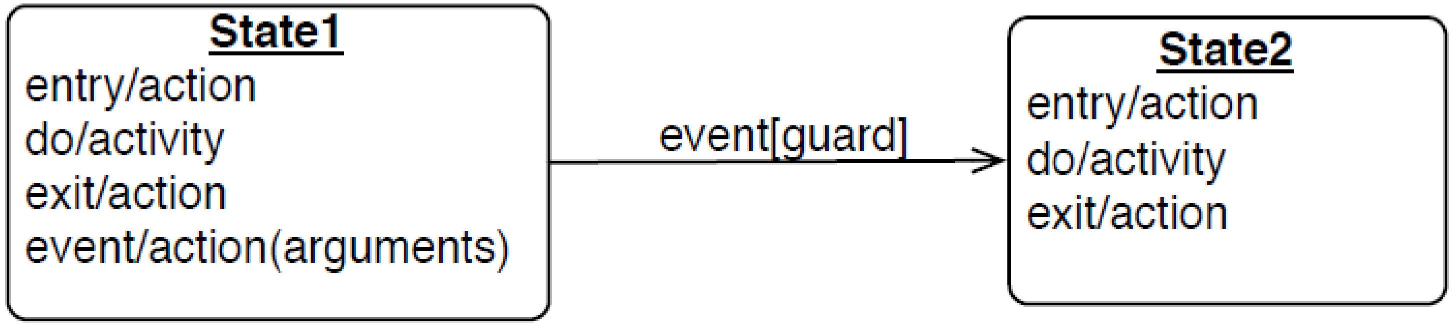
Transitions and Conditions

- A transition is an instantaneous change from one state to another.

- UML notation for transition is:

A **line** drawn from the **origin state** to the **target state**.



Transitions and Conditions

- A transition is said to fire upon the change from the **source state** to the **target state**.
  - Ex: when a called phone is answered, the phone line transitions from the Ringing state to Connected State.

- A **guard condition** is a Boolean expression that must be true in order for a transition to occur.

- The syntax for guard and event along a transition is :

event[guard]

Transitions and Conditions

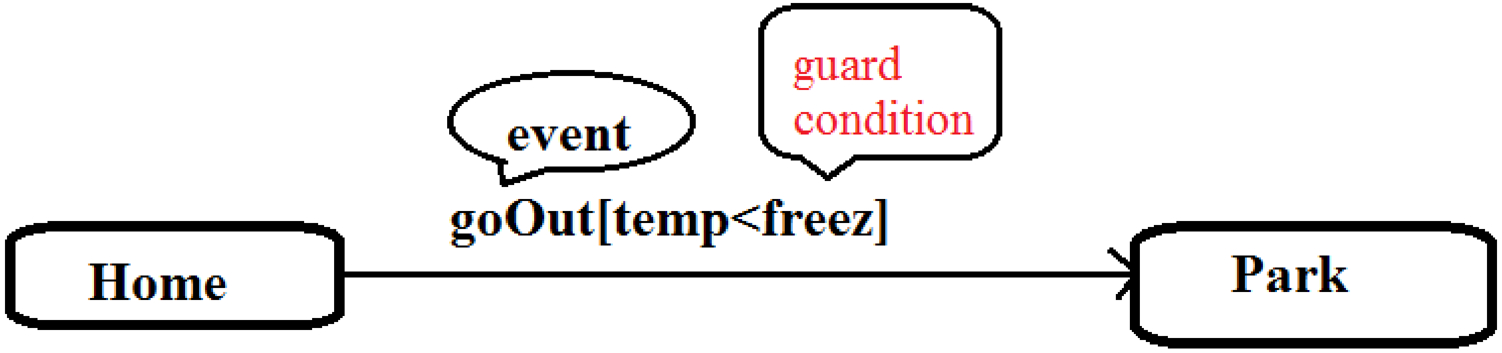
Consider the following example:

Here identify events and states

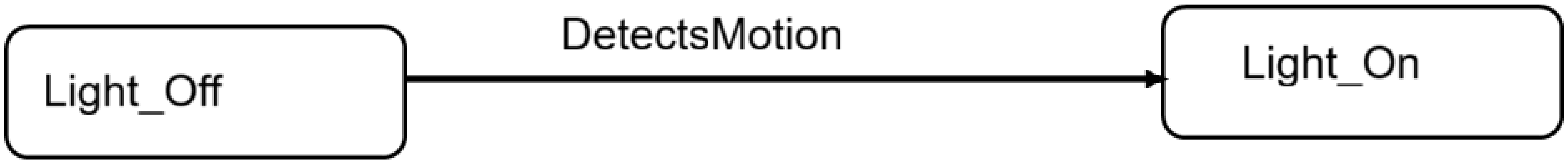
“When you go out in the morning, if the temperature is below freezing, then put on your gloves”

Transitions and Conditions

When you go out in the morning (Event),
if the temperature is below freezing (guard condition),
then put on your gloves(action/event).



Statement: "When a sensor detects motion in a room, turn on the lights"



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State Diagram

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State Diagram

- The **state model** consists of multiple state diagrams, one state diagram for each class.

- State diagram can represent
  - **Continuous loop** or
  - **One-shot life cycle**.

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Continuous loop State Diagram

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Example – state diagram for Telephone Line

Consider the class for telephone line with following activities and states:

As a start of a call, the telephone line is idle. When the phone receiver is picked from hook, it gives a dial tone and can accept the dialing of digits. If after getting dial tone, if the user doesn't dial number within time interval then time out occurs and phone line gets idle. After dialing a number, if the number is invalid then some recorded message is played. Upon entry of a valid number, the phone system tries to connect a call & routes it to proper destination. If the called person answers the phone, the conversation can occur. When called person hangs up, the phone disconnects and goes to idle state.

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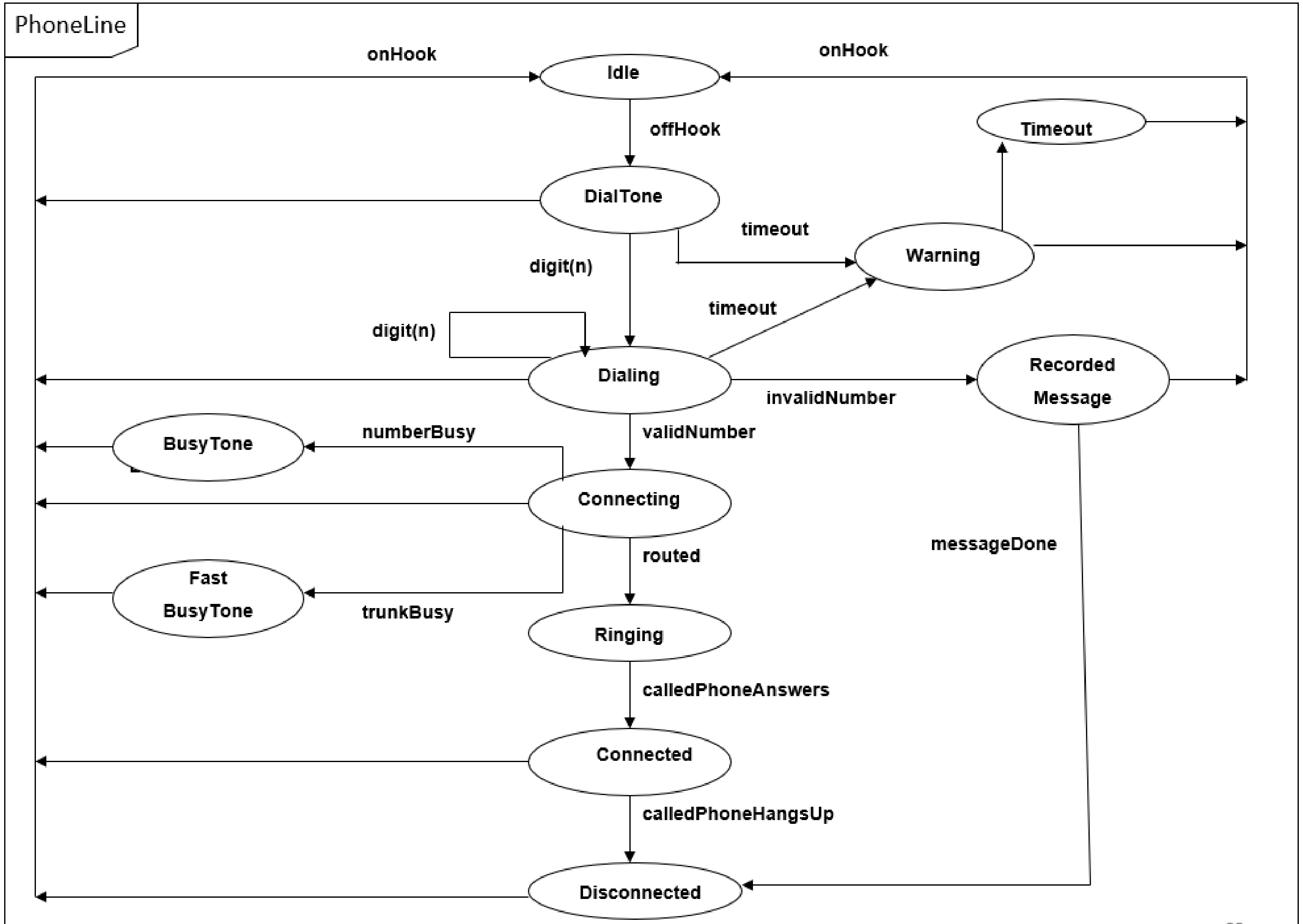
State Diagram – Continuous loop example



State diagram for a telephone line

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One-Shot State Diagram

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One shot State Diagram

- One shot diagrams **represents objects with finite lives** and have **initial** & **final** states.
- A one shot diagram has a **start** - depicted as a **solid circle** - corresponds to **object creation**
- A **final state** is depicted with a **bulls eye** - corresponds to **object destruction**.

In **two ways** we can demonstrate one shot state diagram:

- **with Finite lives**
- **By using Entry and Exit points**

(i) with Finite lives

- One-shot state diagrams represent objects with **finite lives** and have **initial and final states.**
- The **initial state** is entered on **creation** of an object; entry of the final state implies **destruction** of the object.
- The next figure shows a simplified life cycle of a chess game with a default initial state (solid circle) and a default final state (bull's eye)

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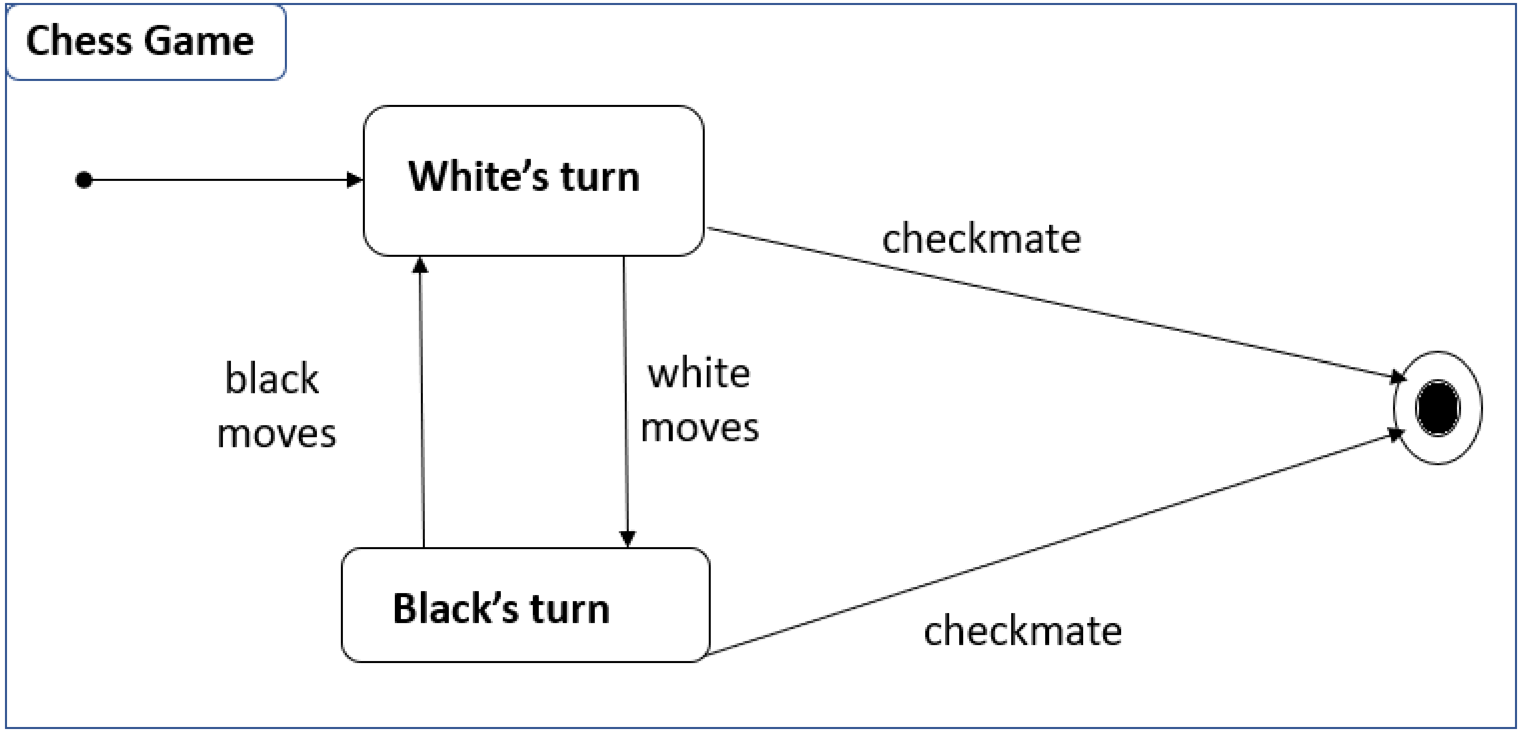
One shot State Diagram

For example:

- The figure shows a **simplified life cycle** of a **chess game** with a default initial state (solid circle) and a default final state (bull's eye)



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(ii) Entry and exit points

- As an alternate notation, you can indicate **initial and final states** via entry and exit points.
- In the next figure, the *start* entry point leads to **white's first** turn, and the chess game eventually ends with **one of three** possible outcomes.
- Entry points (hollow circles) and exit points (circles enclosing an "x") appear on the state diagram's perimeter and may be named.

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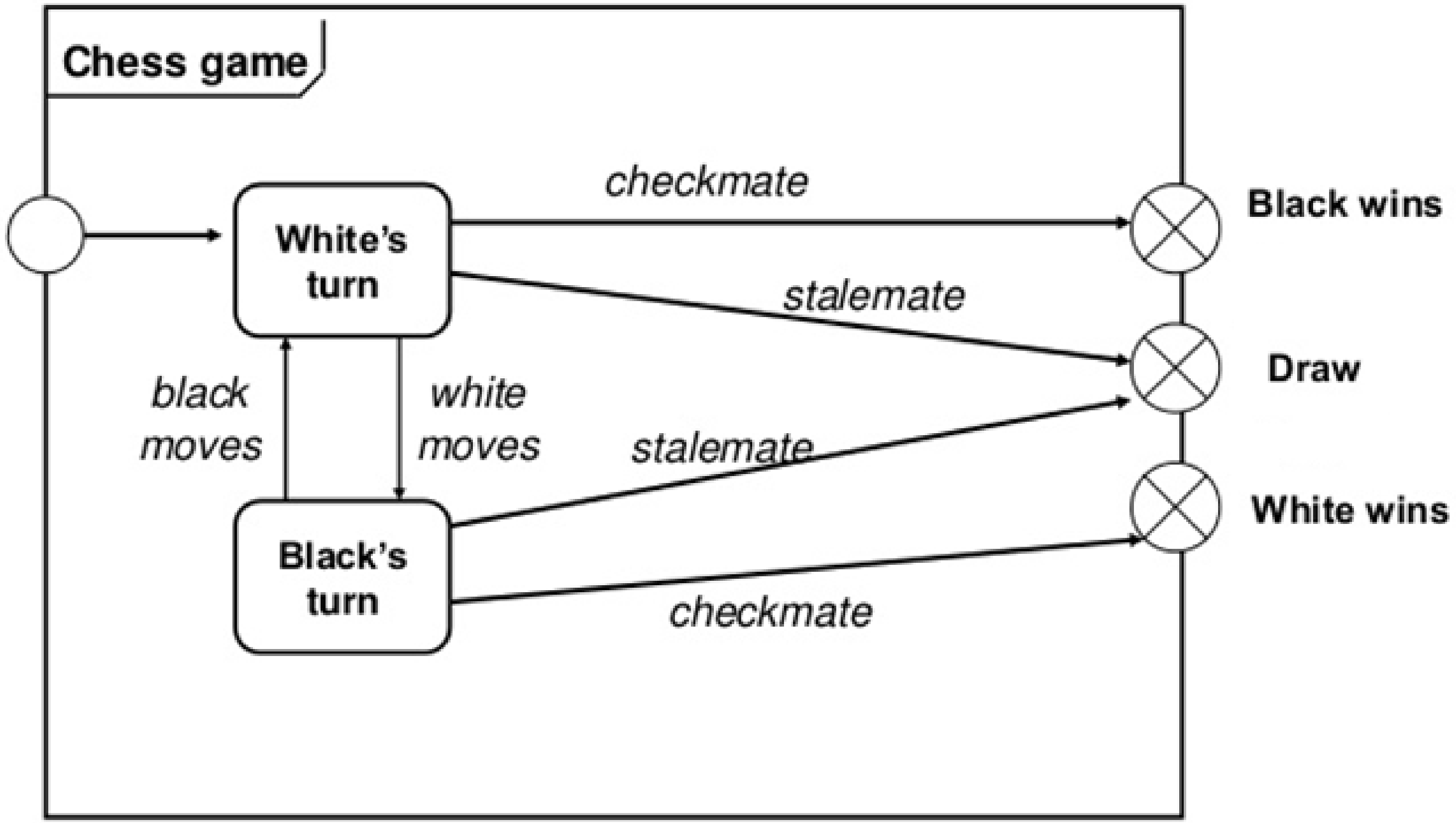
One Shot State Diagram

Example - entry and exit points



Try this: For the Given scenario, identify the events that causes the change of state to takes place and draw a state transition diagram

Account Object

An account starts in the "Active" state.
If there are suspicious activities or violation of terms, the account can be "Blocked.“ From the "Blocked" state, the account may be further escalated to the "Suspended" state for severe violations.
If issues are resolved or penalties lifted, the account can be restored to the "Active" state from either "Blocked" or "Suspended."

THANK YOU

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